

RAHUL BELEDE

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Professional Summary

Data Engineer with 6 years of experience designing, deploying, and optimizing large-scale, real-time data platforms. Skilled in Kafka, Python, SQL, and AWS services. Experienced in building high-throughput ETL, streaming pipelines, and containerized microservices that enable low-latency analytics and support production-grade applications.

Technical Skills

- **Languages:** Python, SQL, Bash, C, C++, R, Java, JavaScript, TypeScript, GoLang
- **Cloud:** AWS (S3, Lambda, SQS, SNS, Secrets Manager, DynamoDB, EC2, Fargate, Redshift, Athena, Load Balancer), GCP (BigQuery, Dataflow, Pub/Sub)
- **Data Engineering:** Kafka, Spark, Airflow, Flink, ETL/ELT, Stream Processing, Real-time Analytics
- **DevOps / Platform:** Docker, Kubernetes, Terraform, Jenkins, GitLab CI, GitHub Actions, ArgoCD, Ansible
- **Databases:** PostgreSQL, MySQL, DynamoDB, Redshift, MongoDB, Cassandra, Redis
- **Monitoring:** Prometheus, Grafana, ELK Stack, Datadog, CloudWatch, OpenTelemetry
- **Frameworks:** Spring Boot, Flask, FastAPI, Node.js, Express.js, React
- **Messaging:** Kafka, RabbitMQ, Kinesis, SQS, Pulsar
- **Visualization:** Tableau, Power BI

Experience

Capital One

Feb 2025 – Present

Data Pipeline Tech (Contract) — Banking / Real-Time Analytics

United States

- Engineered real-time data pipelines on AWS using Kafka, Lambda, and Fargate, processing 100M+ events/day.
- Deployed containerized microservices with Kubernetes, enabling scalable production workloads.
- Automated CI/CD pipelines, reducing release cycles by 30% and rollback times by 50%.
- Designed monitoring dashboards and alerting systems, enhancing SLA visibility and reducing detection time.
- Collaborated with analytics and ML teams to deliver real-time datasets for fraud detection, personalization, and reporting.

Ernst & Young (EY)

Jul 2022 – Aug 2023

Consultant 1 — AdTech / Campaign Analytics

India

- Constructed high-volume ETL pipelines with Python and AWS for campaign performance analytics.
- Optimized Kafka and SQS throughput, reducing ingestion latency by 25%.
- Implemented infrastructure automation using Terraform and Kubernetes for consistent environments.
- Delivered real-time analytics datasets to Redshift and S3 for 10+ marketing and analytics teams.
- Integrated distributed logging and alerting, cutting MTTR by 40%.

Accenture

Jul 2021 – Jul 2022

Application Development Analyst — Enterprise Systems / Data Platform Modernization

India

- Developed batch processing and backend services using Python and SQL for financial reporting workflows.
- Migrated ETL workloads to AWS, reducing compute costs by 20% and improving scalability.
- Orchestrated containerized services using Docker and Kubernetes for reliable deployments.
- Monitored deployments with CloudWatch and Prometheus, decreasing deployment failures by 35%.
- Configured Terraform-based infrastructure as code for repeatable cloud environments.

Infosys

Feb 2019 – Jul 2021

Senior Systems Engineer — Telecom / Data Warehousing

India

- Maintained batch and streaming data pipelines for telecom usage and customer records.
- Automated validation tasks using Python scripts, saving 15+ hours/week.
- Migrated legacy workflows to AWS using MySQL, Redshift, and S3.
- Enhanced pipeline reliability with fault-tolerant logic, achieving 99.9% uptime.
- Standardized documentation and onboarding, mentoring 5+ new engineers.

Think and Learn

May 2018 – Dec 2018

Internship — REST API Development

India

- Developed REST APIs using Node.js and optimized MongoDB queries.
- Implemented test automation and performance tuning for backend services.

Projects

- **Event-Driven Microservices Platform — 2024:** Built platform handling 50K+ events/min with saga pattern orchestration. Integrated API gateway for routing and observability.
- **Real-Time Analytics Data Lake — 2024:** Engineered streaming-first data lake ingesting 10TB+ daily. Orchestrated pipelines with Airflow and Kafka, ensuring sub-hourly data availability.
- **Real-Time Fraud Detection Pipeline — 2024:** Implemented Kafka-based ingestion and Lambda-driven event processing for transaction streams. Triggered alerts via SNS and stored processed data in Redshift.
- **Batch-to-Streaming Migration with Airflow — 2024:** Migrated batch ETL to streaming pipelines, reducing data freshness lag from 24 hours to 5 minutes. Managed infrastructure via Terraform.
- **Marketing Attribution Pipeline — 2023:** Modeled customer journeys and calculated multi-touch attribution. Delivered insights via Power BI dashboards.

Certifications

- AWS Certified Solutions Architect – Associate

Education

University of North Texas <i>Master of Science in Data Engineering</i>	Aug 2023 – May 2025 <i>Denton, TX</i>
Amrita Vishwa Vidyapeetham University <i>Bachelor of Technology in Computer Science and Engineering</i>	Aug 2014 – May 2018 <i>Kerala, India</i>